

Lab 1: Packet analysis at application layer using Wireshark

SCSR1213 Network Communications

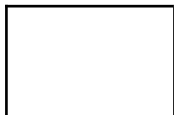
Universiti Teknologi Malaysia

Objective:

1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce student with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

Name : _____ TAN YI YA _____
Metric No : _____ A23CS0187 _____
Section : _____ 01 _____



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PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

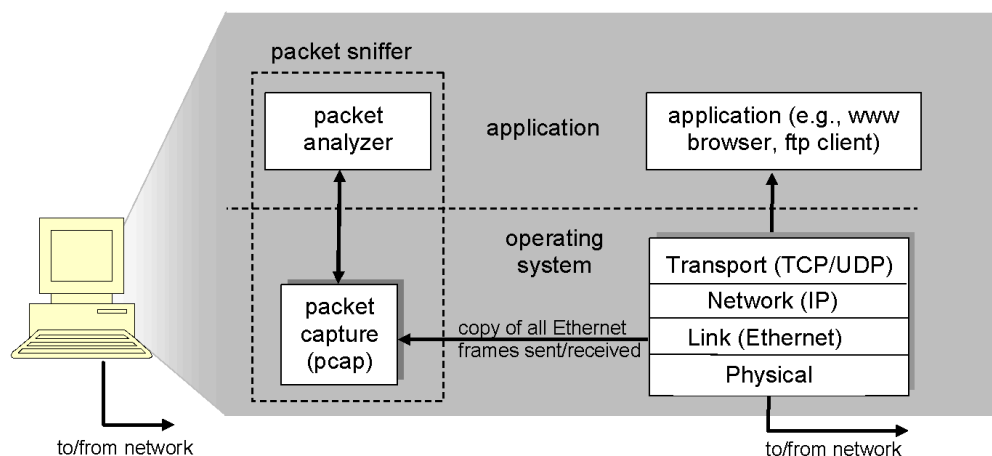


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

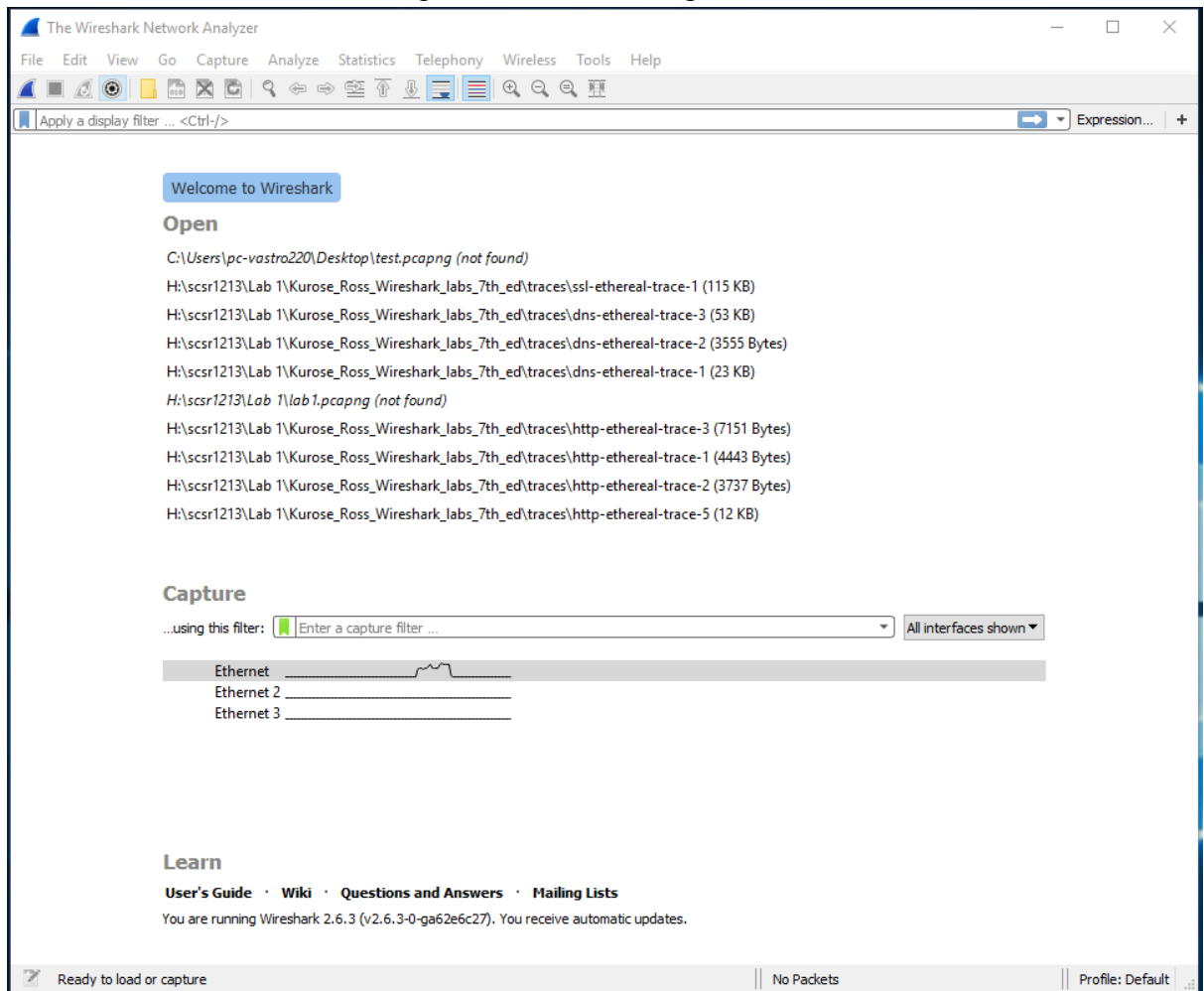


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

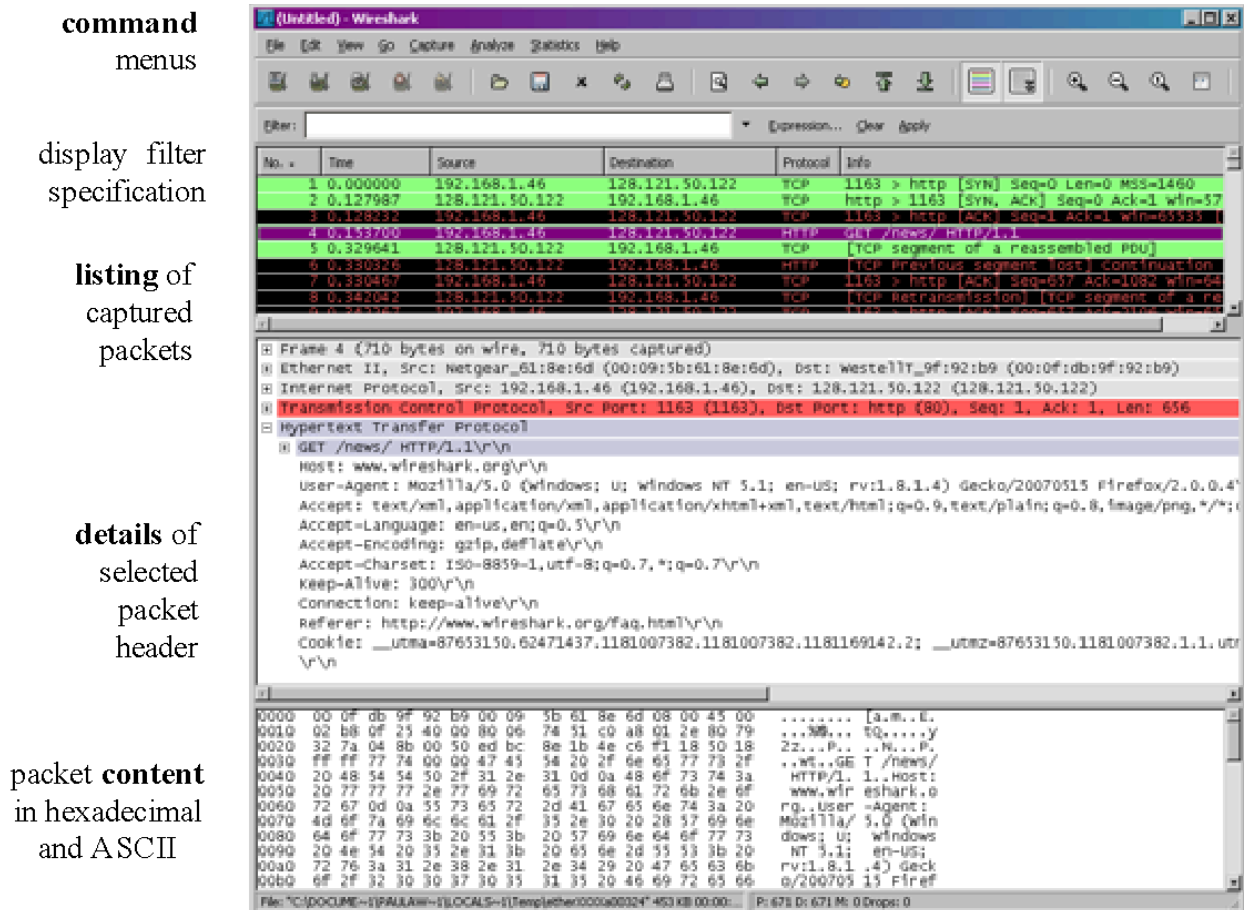


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

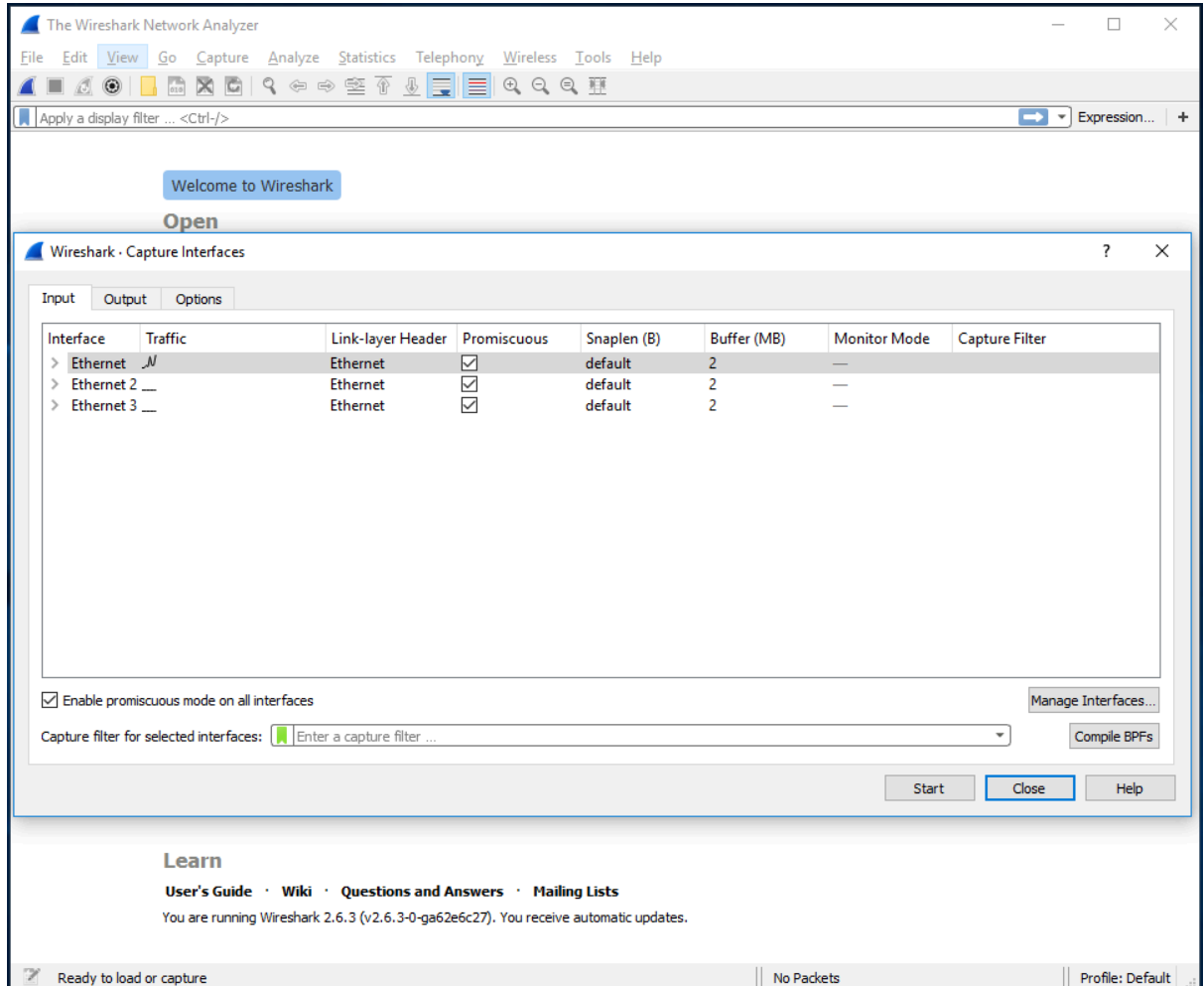


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

The image shows the Wireshark interface with a packet capture in progress. The top pane displays a list of 90 captured packets. The bottom pane shows the detailed view of the first packet (No. 80), which is an ARP request from Dell_87:3b:e4 to the broadcast address 255.255.255.255.

No.	Time	Source	Destination	Protocol	Length	Info
80	4.169920	Dell_87:3b:e4	Broadcast	ARP	60	Who has 10.60.83.65? Tell 10.60.80.70
81	4.344789	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
82	4.346792	10.60.82.204	255.255.255.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
83	4.347075	10.60.82.204	10.60.83.255	DB-LSP...	255	Dropbox LAN sync Discovery Protocol
84	4.351986	HewlettP_15:15:2b	Broadcast	ARP	60	Who has 10.60.80.15? Tell 10.60.80.167
85	4.479278	Cisco_0d:7b:04	Spanning-tree-(for-...	STP	119	MST. Root = 0/10/00:23:04:ee:be:03 Cost = 20004 Port
86	4.530418	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
87	4.530808	10.60.83.171	239.255.255.250	SSDP	216	M-SEARCH * HTTP/1.1
88	4.700996	Dell_75:07:b3	Broadcast	ARP	60	Who has 10.60.80.162? Tell 10.60.80.64
89	4.767952	10.60.82.204	10.60.83.255	NBNS	92	Name query NB MSI-PC<1c>
90	4.995012	10.60.80.78	255.255.255.255	DB-LSP...	176	Dropbox LAN sync Discovery Protocol

Frame 1: 107 bytes on wire (856 bits), 107 bytes captured (856 bits) on interface 0
 Ethernet II, Src: Dell_23:e3:3a (00:24:e8:23:e3:3a), Dst: Cisco_d5:79:ff (00:14:6a:d5:79:ff)
 Internet Protocol Version 4, Src: 10.60.82.216, Dst: 52.230.84.0
 Transmission Control Protocol, Src Port: 49724, Dst Port: 443, Seq: 1, Ack: 1, Len: 53
 Secure Sockets Layer

0000 00 14 6a d5 79 ff 00 24 e8 23 e3 3a 08 00 45 00 ..j.y..\$.#...E.
 0010 00 5d 64 e9 40 00 80 06 00 00 0a 3c 52 d8 34 e6]d.@...<R.4.
 0020 54 00 c2 3c 01 bb 11 3f 6d 43 3e d4 b9 24 50 18 T<<...?mC>...\$P.
 0030 01 01 e6 49 00 00 17 03 01 00 30 07 75 ee af 3c ...I....0.u...<
 0040 42 81 47 9e 52 29 c3 be ea 2e 88 ac 28 ed bf e6 B.G.R)...(...
 0050 20 36 00 a7 ad f6 ff 8a b0 6d 63 a3 c7 a4 68 ca 6.....mc...h.
 0060 fd 6d b3 05 36 73 0e 3c 54 bc 07 ..m..6s<T..

Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

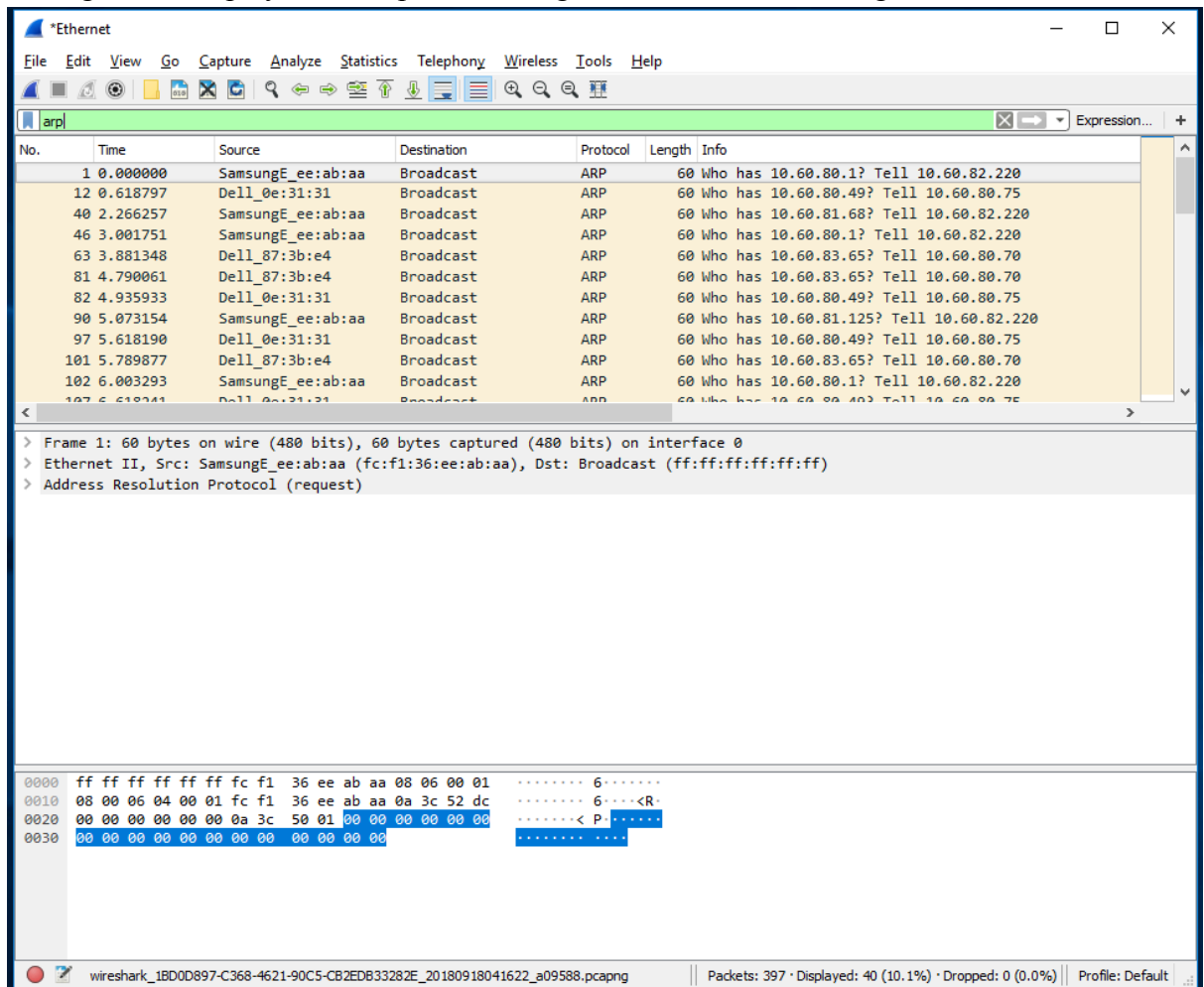


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

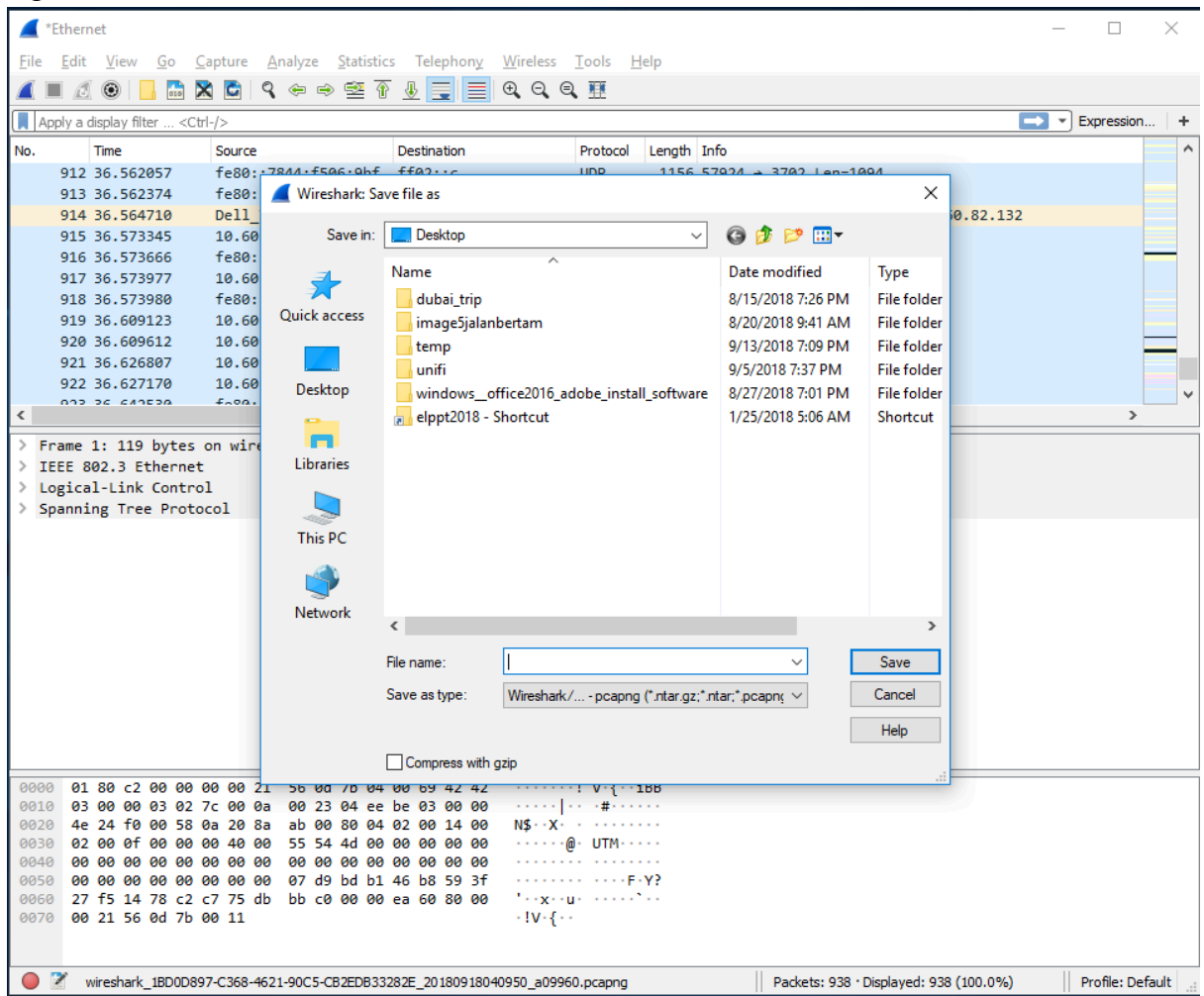


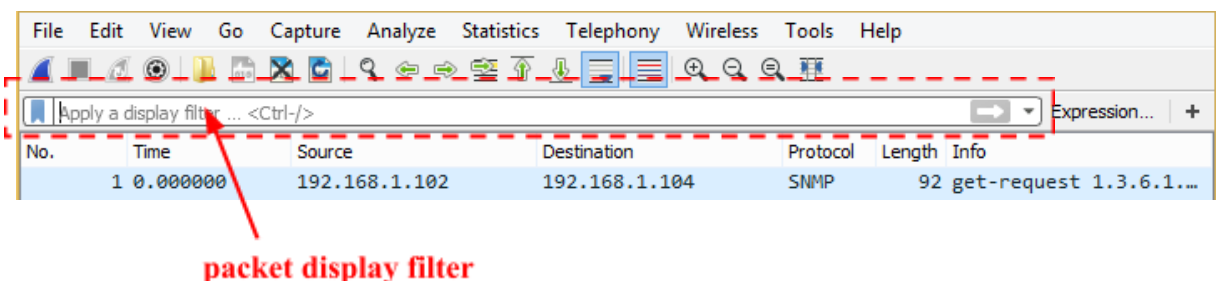
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter “**http**” (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. What version of HTTP is the server running?
[HTTP/1.1](#)
2. What is the IP address of the client computer?
[192.168.1.102](#)
3. What is the IP address of the gaia.cs.umass.edu server?
[128.119.245.12](#)
4. How many bytes of content are being returned to client browser?
[73 bytes](#)
5. What is the status code returned from the server to client browser?
[200 OK for the first request and 404 Not Found for the second request.](#)

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

No, there is no “IF-MODIFIED-SINCE”

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

Yes. The server responded with 200 OK, signifying it has found the file. We can also see the detail section and find the Line-based text data (10 lines) returned.

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

Yes, If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

The returned status code was 304 Not Modified. No, the server does not explicitly return the contents of the file. We know that the file has not been modified since the last HTTP GET request, hence the file was not sent again but cache is used to open the contents of the file.

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

3

2. To which Internet addresses were these GET requests sent?

128.119.245.12

165.193.123.218

134.241.6.82

3. any bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?

3357 bytes

4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

15642 bytes

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2

```
C:\WINDOWS\system32\cmd.exe
C:\Users\pc-vastro220>nslookup yahoo.com
Server: ns1.utm.my
Address: 161.139.250.2

Non-authoritative answer:
Name: yahoo.com
Addresses: 2001:4998:58:1836::11
           2001:4998:c:1023::4
           2001:4998:44:41d::4
           2001:4998:44:41d::3
           2001:4998:58:1836::10
           2001:4998:c:1023::5
           72.30.35.10
           98.137.246.7
           98.137.246.8
           98.138.219.231
           98.138.219.232
           72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server: jlb.utm.my
Address: 161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result

```
C:\WINDOWS\system32\cmd.exe
C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server: ns1.test.time.net.my
Address: 203.121.16.85

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4001:804::200e
           172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

IPv6 Addresses:

- 2603:1030:c02:8::14
- 2603:1020:201:10::10f
- 2603:1010:3:3::5b
- 2603:1030:b:3::152
- 2603:1030:20e:3::23c

IPv4 Addresses:

- 20.231.239.246
- 20.236.44.162
- 20.70.246.20
- 20.76.201.171
- 20.112.250.133

```
Microsoft Windows [Version 10.0.22631.4751]
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Admin>nslookup microsoft.com
```

```
Server:  dns.tm.net.my
```

```
Address: 2001:e68::b:68
```

```
Non-authoritative answer:
```

```
Name:    microsoft.com
```

```
Addresses: 2603:1030:c02:8::14
```

```
           2603:1020:201:10::10f
```

```
           2603:1010:3:3::5b
```

```
           2603:1030:b:3::152
```

```
           2603:1030:20e:3::23c
```

```
           20.231.239.246
```

```
           20.236.44.162
```

```
           20.70.246.20
```

```
           20.76.201.171
```

```
           20.112.250.133
```

```
C:\Users\Admin>
```

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

```
C:\Users\Admin>nslookup -type=ns microsoft.com
Server:  dns.tm.net.my
Address:  2001:e68::b:68

Non-authoritative answer:
microsoft.com    nameserver = ns2-39.azure-dns.net
microsoft.com    nameserver = ns3-39.azure-dns.org
microsoft.com    nameserver = ns4-39.azure-dns.info
microsoft.com    nameserver = ns1-39.azure-dns.com

ns1-39.azure-dns.com    internet address = 150.171.10.39
ns1-39.azure-dns.com    AAAA IPv6 address = 2603:1061:0:10::27
ns2-39.azure-dns.net    internet address = 150.171.16.39
ns2-39.azure-dns.net    AAAA IPv6 address = 2620:1ec:8ec:10::27
ns3-39.azure-dns.org    internet address = 13.107.222.39
ns3-39.azure-dns.org    AAAA IPv6 address = 2a01:111:4000:10::27
ns4-39.azure-dns.info    internet address = 13.107.206.39
ns4-39.azure-dns.info    AAAA IPv6 address = 2620:1ec:bda:10::27

C:\Users\Admin>
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all

```
C:\WINDOWS\system32\cmd.exe
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-AA-18-73-0D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : utm.my
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : 00-24-E8-23-E3-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddf4(Preferred)
    IPv4 Address. . . . . : 10.60.82.216(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Tuesday, September 18, 2018 6:08:09 AM
    Lease Expires . . . . . : Thursday, September 20, 2018 6:08:10 AM
    Default Gateway . . . . . : 10.60.80.1
    DHCP Server . . . . . : 10.60.80.1
    DHCPv6 IAID . . . . . : 33563880
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
    DNS Servers . . . . . : 161.139.250.2
                          161.139.168.168
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    Description . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
Command: ipconfig /displaydns

```
C:\WINDOWS\system32\cmd.exe
Record Name . . . . . : wpad.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 5917
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.163.188

-----
Record Name . . . . . : aimsweb.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16483
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.18.215

-----
Record Name . . . . . : my.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16484
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.20.240

C:\Users\pc-vastro220>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.
Command: ipconfig /flushdns


```

C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\pc-vastro220>

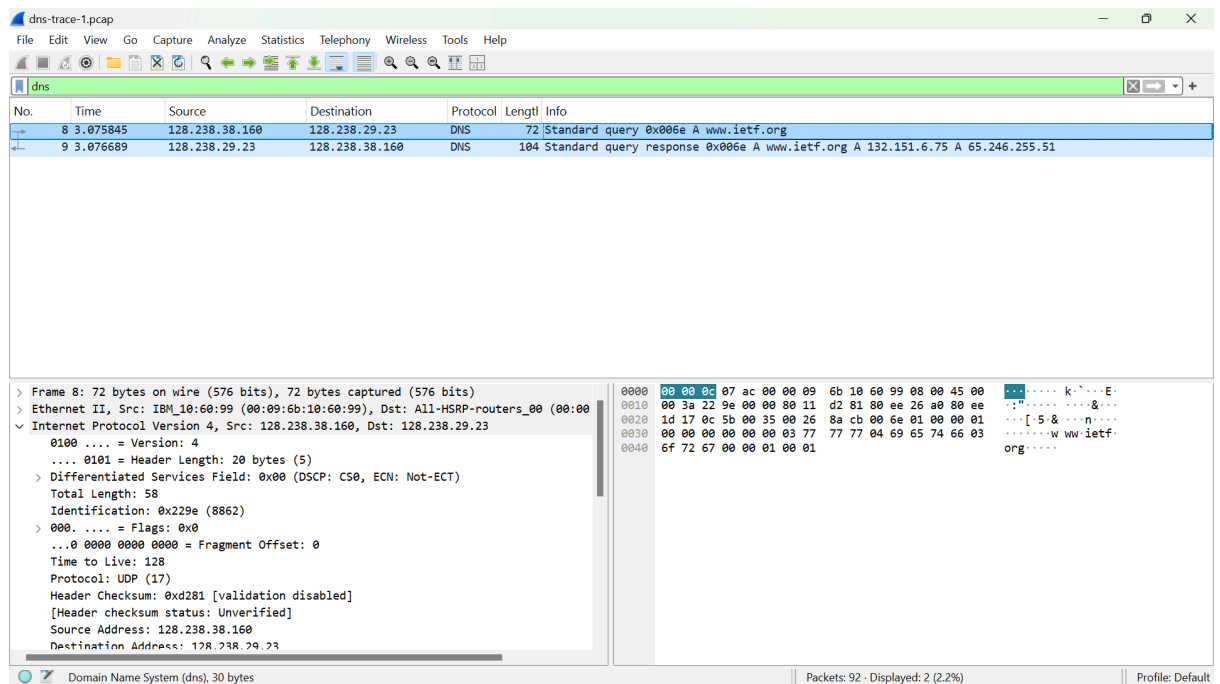
```

Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1. Answer the following questions.
1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

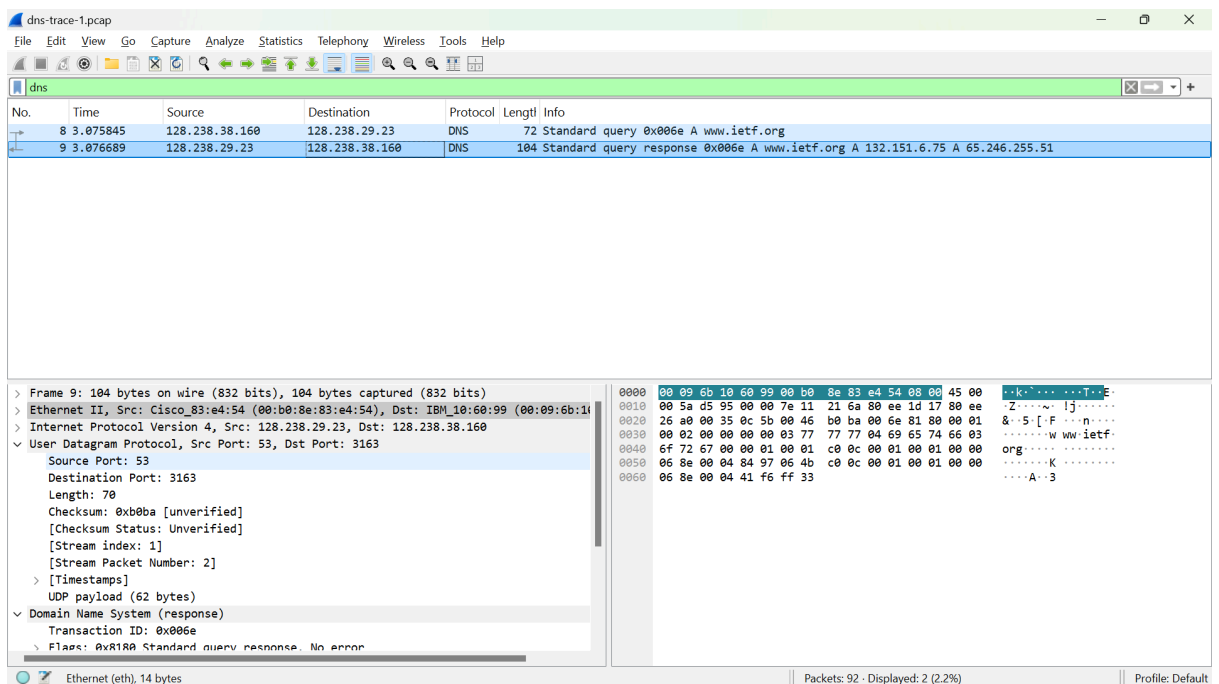
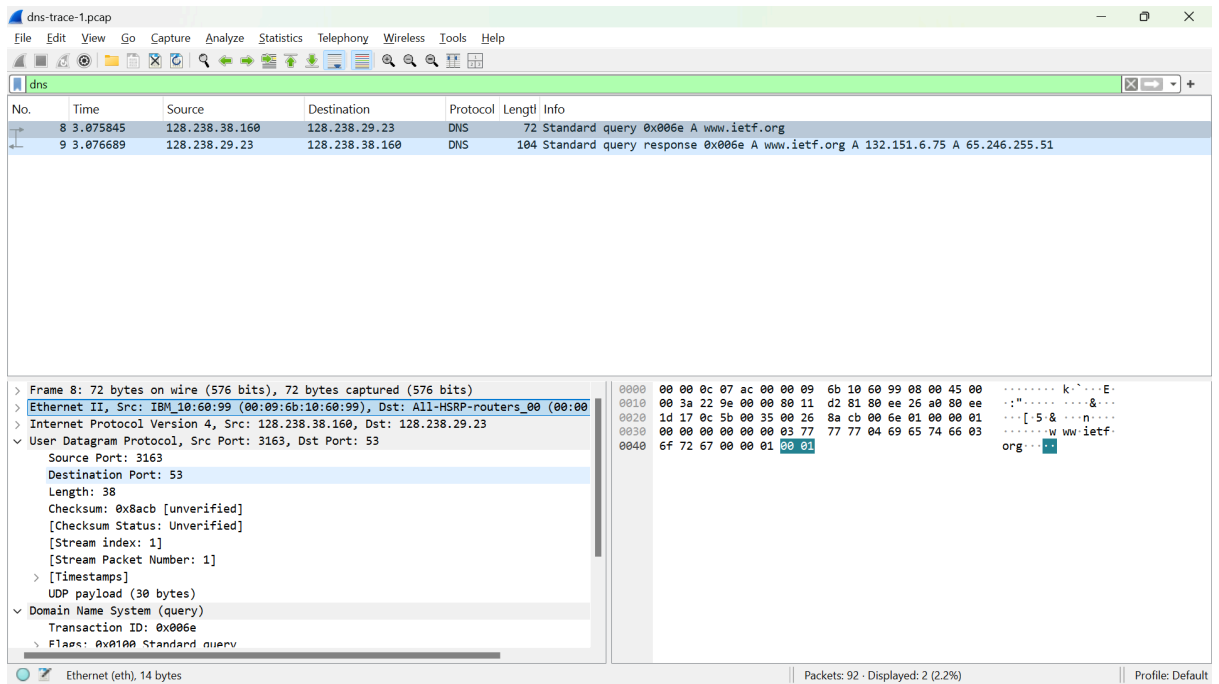
UDP



2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

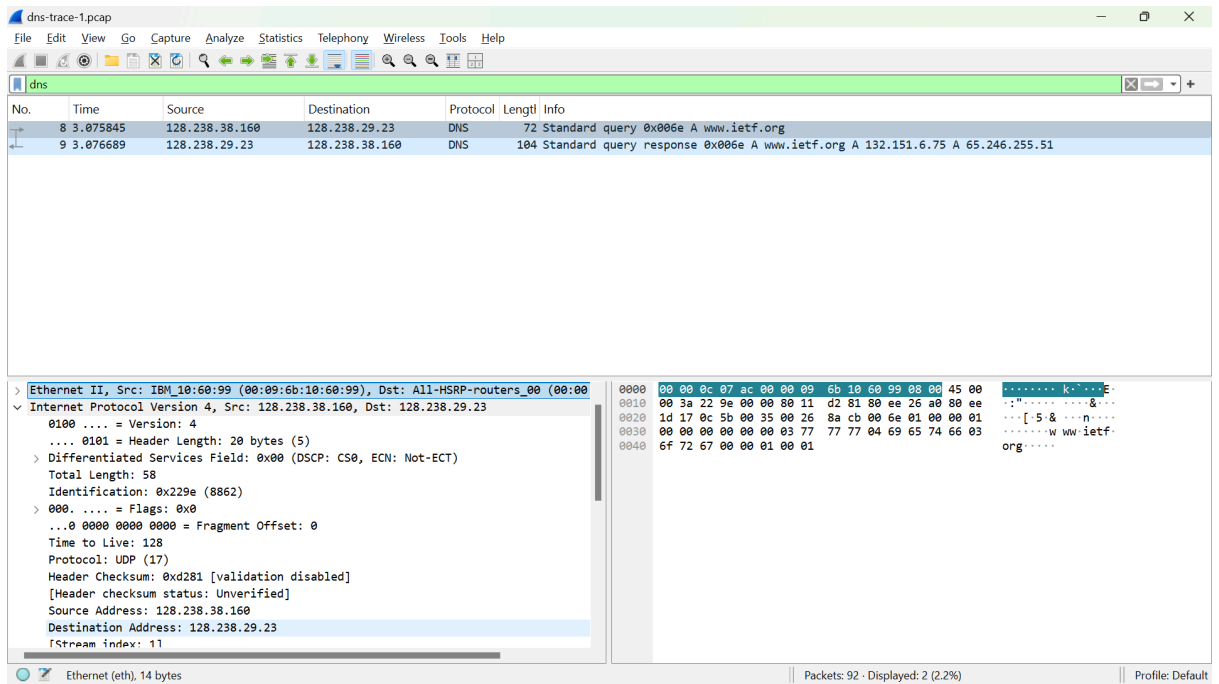
Destination port of DNS query message: 53

Source port of DNS response message: 53



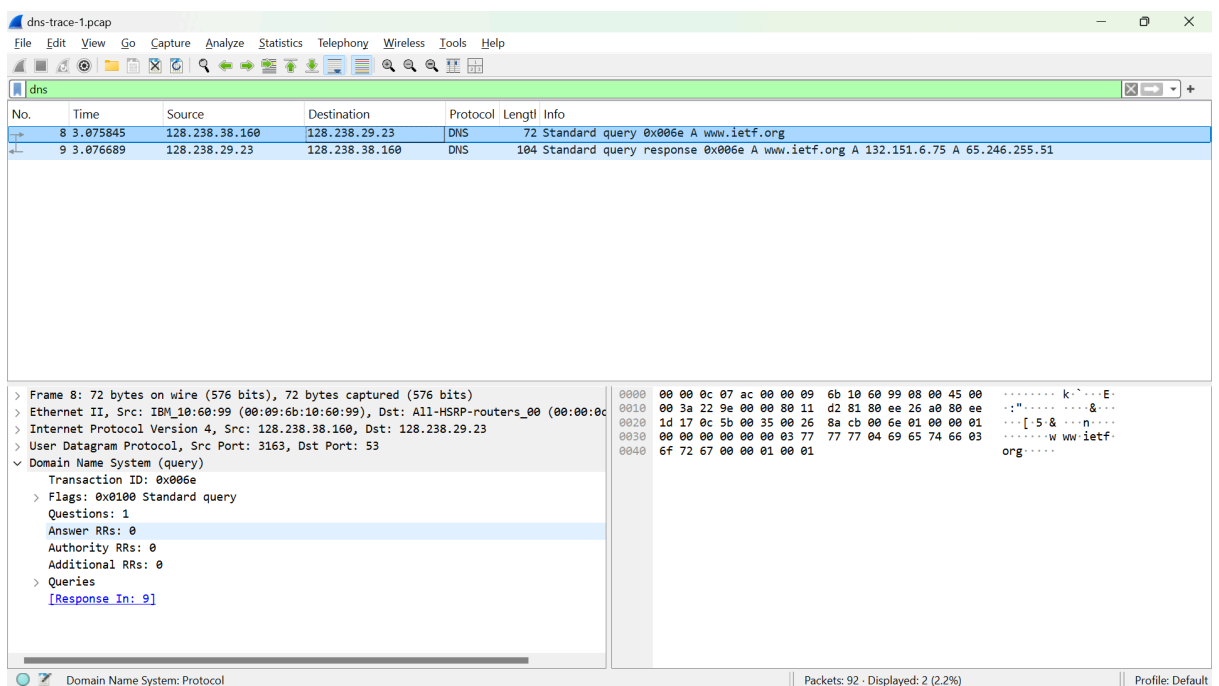
3. To what IP address is the DNS query message sent? Add screenshots in your answer.

Destination IP address: 128.238.29.23



4. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.
- Standard query

No, Answer RRs = 0



5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

2 Answers were provided. it contains the ip address to www.ietf.org

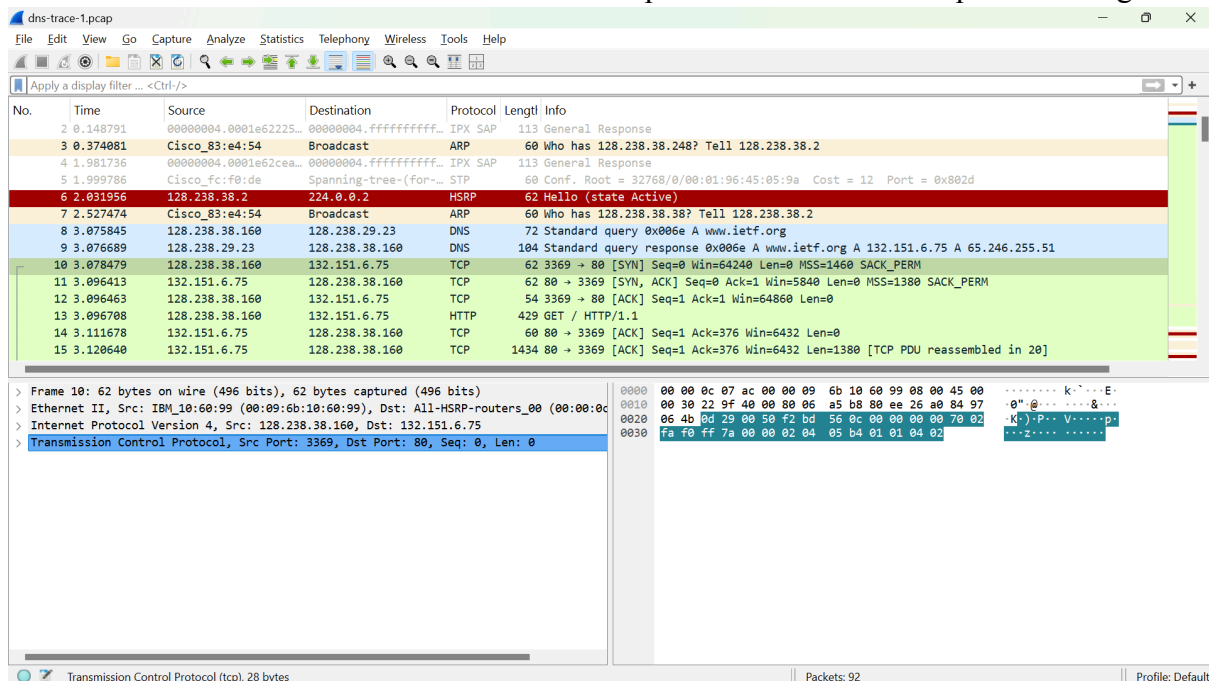
The screenshot shows a Wireshark packet capture of a DNS response. The packet list at the top shows two packets: a query (No. 8) and a response (No. 9). The response packet is selected, and the packet details pane on the left shows the hierarchy: Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (response). The DNS response details are expanded, showing the Transaction ID (0x006e), Flags (Standard query response, No error), Questions (1), Answer RRs (2), Authority RRs (0), and Additional RRs (0). The Answers section is expanded, showing two records for www.ietf.org: one with IP address 132.151.6.75 and another with IP address 65.246.255.51. The packet bytes pane on the right shows the raw data of the DNS response, with the answer section highlighted in blue.

No.	Time	Source	Destination	Protocol	Length	Info
8	3.075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
9	3.076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51

Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
User Datagram Protocol, Src Port: 53, Dst Port: 5163
Domain Name System (response)
Transaction ID: 0x006e
Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 2
Authority RRs: 0
Additional RRs: 0
Queries
Answers
www.ietf.org: type A, class IN, addr 132.151.6.75
www.ietf.org: type A, class IN, addr 65.246.255.51
[Request In: 8]
[Time: 0.000844000 seconds]

0000 00 09 6b 10 60 99 00 b0 8e 83 e4 54 08 00 45 00 ...k...T...E...
0010 00 5a d5 95 00 00 7e 11 21 6a 80 ee 1d 17 80 ee ...Z...l...j...
0020 26 a0 00 35 0c 5b 00 46 b0 ba 00 6e 81 80 00 01 ...&...5...F...n...
0030 00 02 00 00 00 03 77 77 77 04 69 65 74 66 03 ...w...w...w...
0040 6f 72 67 00 00 01 00 01 c0 0c 00 01 00 01 00 00 ...org...K...
0050 06 8e 00 04 84 97 06 4b c0 0c 00 01 00 01 00 00 ...A...3...
0060 06 8e 00 04 41 f6 ff 33

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.
- source IP is the same as the destination IP provided in the DNS response message.



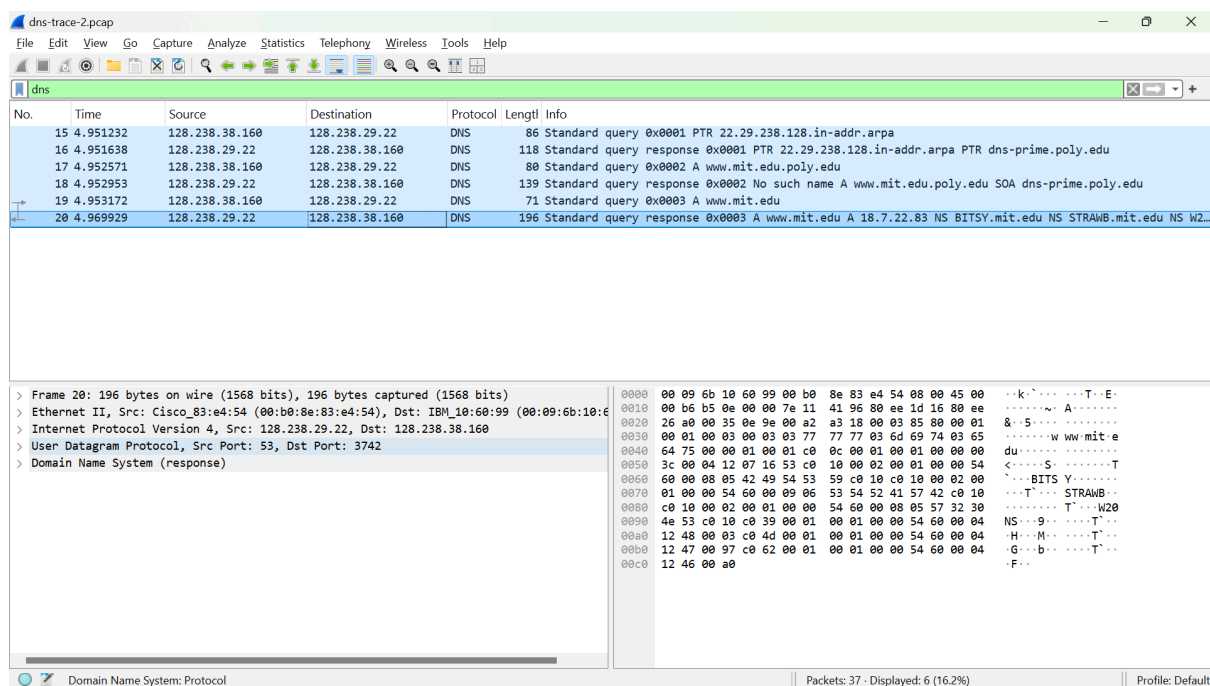
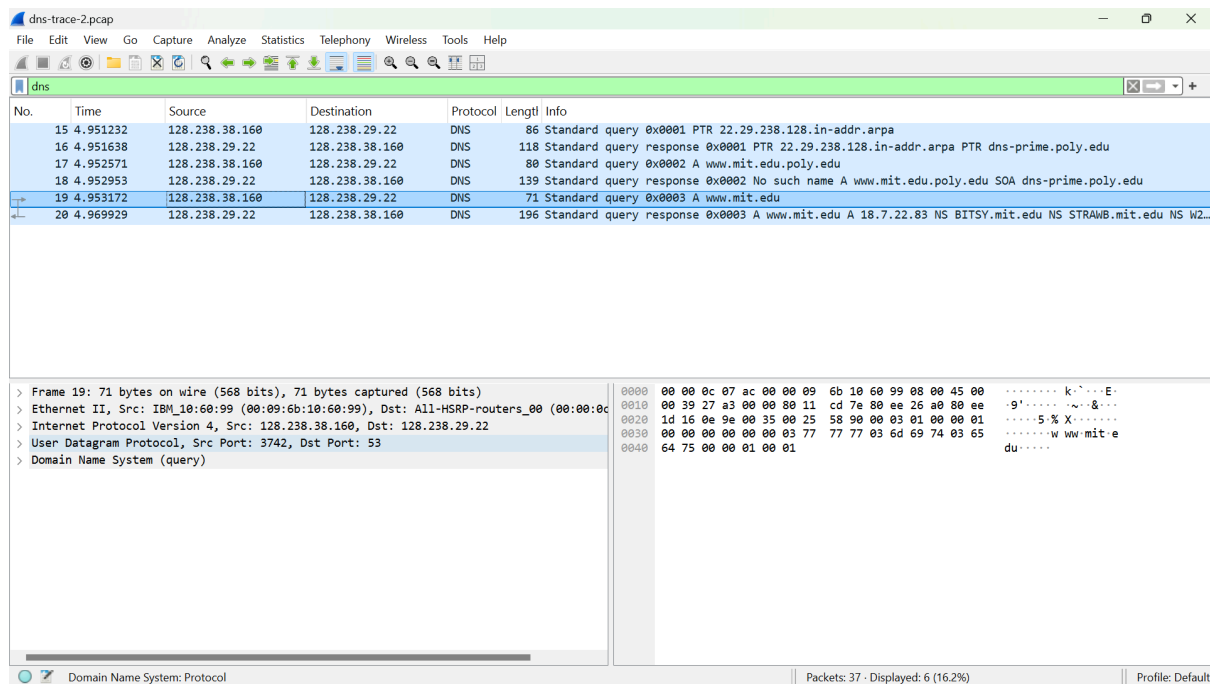
7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?
- No. There is only 2 DNS queries in total, which is the DNS request and respond just now.

- Open packet trace file dns-trace-2 for nslookup.
- We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
- Answer the following questions.

8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

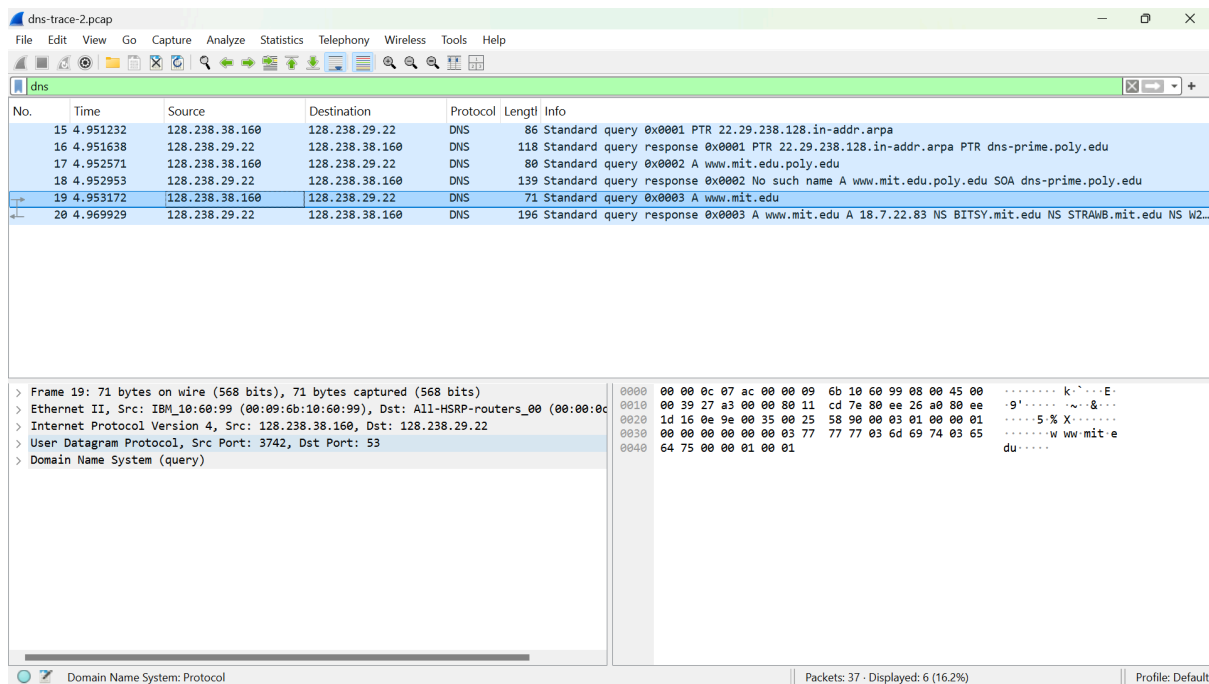
Destination Port for DNS query message: 53

Source Port of DNS response message: 53

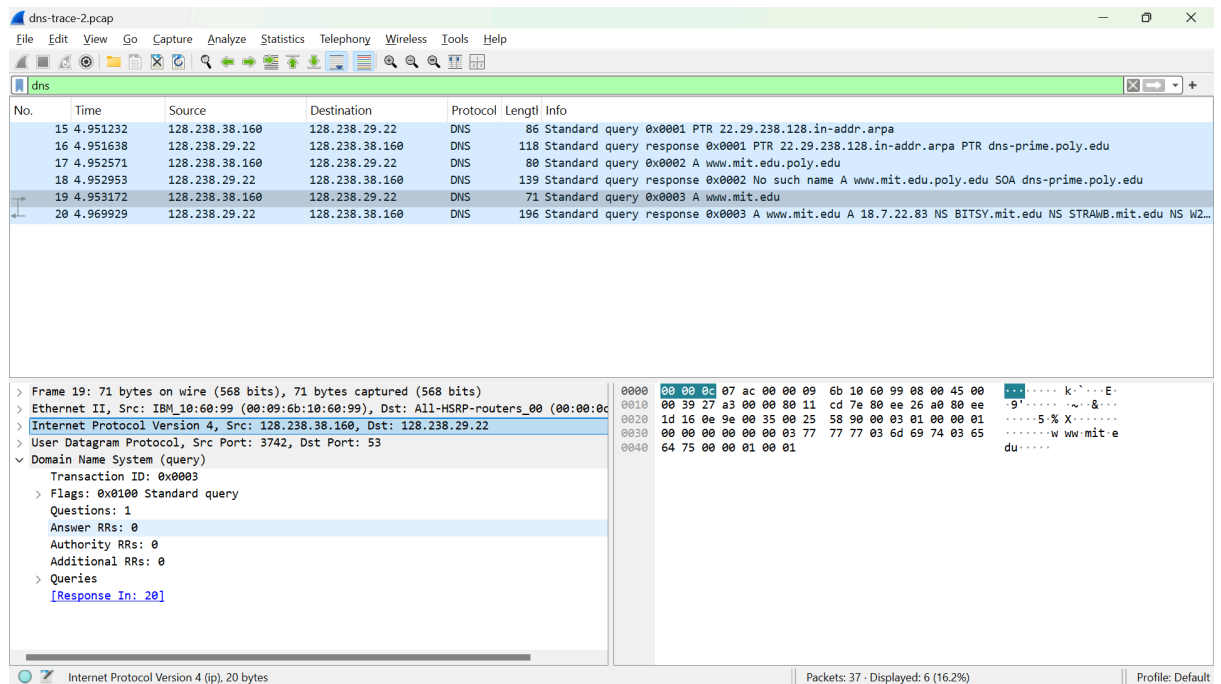


9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.

destination IP address of DNS query message: 128.238.29.22
 No, it is not the IP address of the default local DNS server.



10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.
 Standard query, it does not contain answers.



11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

7 answers (RRs) are provided.

Answers

- www.mit.edu: type A, class IN, addr 18.7.22.83

Authoritative nameservers

- mit.edu: type NS, class IN, ns BITSY.mit.edu
- mit.edu: type NS, class IN, ns STRAWB.mit.edu
- mit.edu: type NS, class IN, ns W20NS.mit.edu

Additional records

- BITSY.mit.edu: type A, class IN, addr 18.72.0.3
- STRAWB.mit.edu: type A, class IN, addr 18.71.0.151
- W20NS.mit.edu: type A, class IN, addr 18.70.0.160

The screenshot displays a Wireshark capture of a DNS response. The packet list at the top shows a standard query response (Frame 20) from 128.238.29.22 to 128.238.38.160. The packet details pane on the left shows the transaction ID 0x0003, flags 0x8580, and a standard query response with no error. The packet bytes pane on the right shows the raw data of the DNS response, including the header and the answer section with 7 records: 1 A record for www.mit.edu (18.7.22.83) and 3 NS records for mit.edu (BITSY.mit.edu, STRAWB.mit.edu, W20NS.mit.edu).

dns-trace-2.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAMB.mit.edu NS W2...

Authority RRs: 3
Additional RRs: 3

> Queries

✓ Answers

> www.mit.edu: type A, class IN, addr 18.7.22.83

✓ Authoritative nameservers

> mit.edu: type NS, class IN, ns BITSY.mit.edu

> mit.edu: type NS, class IN, ns STRAMB.mit.edu

> mit.edu: type NS, class IN, ns W20NS.mit.edu

✓ Additional records

> BITSY.mit.edu: type A, class IN, addr 18.72.0.3

> STRAMB.mit.edu: type A, class IN, addr 18.71.0.151

> W20NS.mit.edu: type A, class IN, addr 18.70.0.160

[Request In: 19]
[Time: 0.016757000 seconds]

0000 00 02 05 10 60 99 00 b0 8e 83 e4 54 08 00 45 00T..E..
0010 00 b6 b5 0e 00 00 7e 11 41 9e 80 ee 1d 16 80 eeA.....
0020 26 a0 00 35 0e 9e 00 a2 a3 18 00 03 85 80 00 01S.....
0030 00 01 00 03 00 03 03 77 77 77 03 6d 69 74 03 65www.mit.e
0040 64 75 00 00 01 00 01 c0 0c 00 01 00 01 00 00 00du.....
0050 3c 00 04 12 07 16 53 c0 10 00 02 00 01 00 00 54S.....T
0060 60 00 08 05 42 49 54 53 59 c0 10 c0 10 00 02 00BITS Y.....
0070 01 00 00 54 60 00 09 06 53 54 52 41 57 42 c0 10T.....STRAMB..
0080 c0 10 00 02 00 01 00 00 54 60 00 08 05 57 32 30T.....W20
0090 4e 53 c0 10 c0 39 00 01 00 01 00 00 54 60 00 04NS...9.....T..
00a0 12 48 00 03 c0 4d 00 01 00 01 00 00 54 60 00 04H...M.....T..
00b0 12 47 00 97 c0 62 00 01 00 01 00 00 54 60 00 04G...b.....T..
00c0 12 46 00 a0F...

Internet Protocol Version 4 (ip), 20 bytes

Packets: 37 · Displayed: 6 (16.2%)

Profile: Default